DATA COMMUNICATIONS & NETWORKING: QUIZ 1

Topic: Analog Modulation (AM, FM) & Amplitude Shift Keying (ASK)

Name: _____ Student ID: _____ Score: ____ / 25

Question 1 (Multiple Choice)

Why does an analog baseband signal need to be modulated before transmission through a band-pass channel?

[A] To completely eliminate all frequency components from the source information

[B] To shift the signal spectrum into a frequency band supported by the transmission medium

[C] To convert a continuous waveform directly into discrete binary line coding

[D] To force the carrier frequency to zero Hertz (0 Hz) for direct-current transmission

Question 2 (Multiple Choice)

What is the fundamental engineering principle of carrier modulation?

[A] Maintaining all high-frequency carrier parameters completely constant during transmission

[B] Suppressing the carrier entirely and transmitting only the raw baseband voltage levels

[C] Systematically varying at least one characteristic of a carrier signal according to the modulating information

[D] Converting the continuous carrier wave into a sequence of discrete binary pulse code elements

Question 3 (Calculation: AM Spectrum)

Consider standard Amplitude Modulation (AM). The carrier frequency is $f_c = 200$ kHz, and the modulating audio signal has a maximum frequency of $B_m = 15$ kHz.

- Total required bandwidth (BW_{AM}) = _____ kHz
- Lower sideband frequency (f_{lower}) = _____ kHz
- Upper sideband frequency (f_{upper}) = _____ kHz

Question 4 (Calculation: FM Bandwidth)

An audio signal with a bandwidth of $B_m = 6\ \text{kHz}$ is transmitted using Frequency Modulation (FM). Assuming a modulation index $\beta = 4$ and using Carson's rule where $BW_{\text{FM}} = 2(1+\beta)B_m = 10B_m$:

- Required FM bandwidth (BW_{FM}) = _____ kHz ($=$ _____ Hz)

Question 5 (Calculation: ASK Parameters)

An Amplitude Shift Keying (ASK) signal has an allocated bandwidth spanning from $100\ \text{kHz}$ to $500\ \text{kHz}$. Let the filter roll-off factor be $d = 1$, and bits per baud $r = 1$.

- Total bandwidth (BW) = _____ kHz
- Carrier frequency (f_c) = _____ kHz
- Baud rate (S) = _____ kBaud
- Data rate (N) = _____ kbps

Question 6 (Problem Solving: Full-Duplex ASK)

An ASK communication link operates between $2000\ \text{Hz}$ and $8000\ \text{Hz}$ in full-duplex transmission mode. The total bandwidth is divided equally between the forward and backward directions with no guard band.

- Bandwidth allocated for each direction = _____ kHz
- Carrier frequency for forward channel (f_{c1}) = _____ kHz
- Carrier frequency for backward channel (f_{c2}) = _____ kHz

Question 7 (Calculation: Reverse AM Parameters)

A commercial AM broadcast signal occupies the frequency range from $540\ \text{kHz}$ to $560\ \text{kHz}$.

- Carrier frequency (f_c) = _____ kHz
- Total bandwidth (BW) = _____ kHz
- Maximum frequency of modulating signal (B_m) = _____ kHz

Question 8 (Calculation: FM Bandwidth Expansion)

A baseband speech signal has bandwidth $B_m = 4\ \text{kHz}$. It can be transmitted via either AM or wideband FM with modulation parameter $\beta = 4$.

- Bandwidth needed for AM = _____ kHz
- Bandwidth needed for FM = _____ kHz
- Bandwidth expansion factor ($\frac{BW_{\text{FM}}}{BW_{\text{AM}}}$) = _____

Question 9 (Calculation: ASK Bandwidth vs Data Rate)

We need to send digital data at a bit rate of $N = 64\ \text{kbps}$ using binary ASK ($r = 1$). If the channel filter factor is $d = 0.5$:

- Baud rate (S) = _____ kBaud
- Minimum required channel bandwidth (BW) = _____ kHz

Question 10 (Problem Solving: Full-Duplex ASK with Guard Band)

A full-duplex binary ASK transmission system spans the frequency band from $10\ \text{kHz}$ to $50\ \text{kHz}$. A central guard band of $4\ \text{kHz}$ is reserved between the two directions to prevent cross-talk. The remaining spectrum is split evenly between the forward and backward directions.

- Bandwidth available for each direction = _____ kHz
- Center carrier frequency for forward path (f_{c1}) = _____ kHz
- Center carrier frequency for backward path (f_{c2}) = _____ kHz

Question 11 (Multiple Choice: Modulation Comparison)

Why is Frequency Modulation (FM) generally less susceptible to ambient noise and amplitude distortions than Amplitude Modulation (AM)?

[A] FM signals operate at lower carrier frequencies than AM

[B] Noise primarily affects signal amplitude; an FM receiver can clip amplitude fluctuations without losing information encoded in frequency

[C] FM requires significantly less bandwidth than AM to send identical baseband signals

[D] FM uses digital line coding techniques to correct corrupted bits

Question 12 (Matching / Short Answer)

Identify which carrier characteristic is modulated by the baseband signal for each technique:

1. Amplitude Modulation (AM): _____
2. Frequency Modulation (FM): _____
3. Phase Modulation (PM): _____
4. Amplitude Shift Keying (ASK): _____

Answer Key & Detailed Explanations

Question 1: **[B] To shift the signal spectrum into a frequency band supported by the transmission medium**

Explanation: Band-pass channels do not allow baseband signals (frequencies starting near 0 Hz) to pass through efficiently without severe attenuation. Modulation translates the baseband spectrum to high frequencies matched to channel passbands and physical antenna dimensions[cite: 5].

Question 2: **[C] Systematically varying at least one characteristic of a carrier signal according to the modulating information**

Explanation: A high-frequency sinusoidal carrier has three principal parameters: amplitude, frequency, and phase. Modulation alters one (or a combination) of these parameters in proportion to the input signal[cite: 5].

Question 3: $BW_{\text{AM}} = 30$ kHz; $f_{\text{lower}} = 185$ kHz; $f_{\text{upper}} = 215$ kHz

Mathematical Derivation:

$$\text{Bandwidth: } BW_{\text{AM}} = 2 \times B_m = 2 \times 15 \text{ kHz} = 30 \text{ kHz}$$

$$\text{Lower sideband: } f_{\text{lower}} = f_c - B_m = 200 \text{ kHz} - 15 \text{ kHz} = 185 \text{ kHz}$$

$$\text{Upper sideband: } f_{\text{upper}} = f_c + B_m = 200 \text{ kHz} + 15 \text{ kHz} = 215 \text{ kHz}$$

[cite: 6]

Question 4: $BW_{\text{FM}} = 60$ kHz $(60,000$ Hz$)$

Mathematical Derivation: Using Carson's bandwidth rule with $\beta = 4$:

$$BW_{\text{FM}} = 2(1 + \beta)B_m = 2(1 + 4) \times 6 \text{ kHz} = 10 \times 6 \text{ kHz} = 60 \text{ kHz} = 60,000 \text{ Hz}$$

[cite: 6]

Question 5: $BW = 400\ \text{kHz}$; $f_c = 300\ \text{kHz}$; $S = 200\ \text{kBaud}$; $N = 200\ \text{kbps}$
Mathematical Derivation:

$$BW = f_{\text{high}} - f_{\text{low}} = 500\ \text{kHz} - 100\ \text{kHz} = 400\ \text{kHz}$$

$$f_c = \frac{f_{\text{low}} + f_{\text{high}}}{2} = \frac{100\ \text{kHz} + 500\ \text{kHz}}{2} = 300\ \text{kHz}$$

For ASK, $BW = (1 + d) \times S$. With $d = 1$:

$$400\ \text{kHz} = (1 + 1) \times S \implies S = \frac{400}{2} = 200\ \text{kBaud}$$

Since $r = 1\ \text{bit/baud}$:

$$N = r \times S = 1 \times 200\ \text{kBaud} = 200\ \text{kbps}$$

[cite: 7]

Question 6: **Bandwidth/direction =** $3\ \text{kHz}$; $f_{c1} = 3.5\ \text{kHz}$; $f_{c2} = 6.5\ \text{kHz}$
Mathematical Derivation: Total bandwidth $= 8000\ \text{Hz} - 2000\ \text{Hz} = 6000\ \text{Hz} = 6\ \text{kHz}$.

$$BW_{\text{direction}} = \frac{6\ \text{kHz}}{2} = 3\ \text{kHz} = 3000\ \text{Hz}$$

Forward band: $2000\ \text{Hz}$ to $5000\ \text{Hz} \implies f_{c1} = \frac{2000+5000}{2} = 3500\ \text{Hz} = 3.5\ \text{kHz}$.
Backward band: $5000\ \text{Hz}$ to $8000\ \text{Hz} \implies f_{c2} = \frac{5000+8000}{2} = 6500\ \text{Hz} = 6.5\ \text{kHz}$[cite: 7].

Question 7: $f_c = 550\ \text{kHz}$; $BW = 20\ \text{kHz}$; $B_m = 10\ \text{kHz}$
Mathematical Derivation:

$$BW = 560\ \text{kHz} - 540\ \text{kHz} = 20\ \text{kHz}$$

$$f_c = \frac{540 + 560}{2} = 550\ \text{kHz}$$

$$BW_{\text{AM}} = 2B_m \implies B_m = \frac{BW}{2} = \frac{20\ \text{kHz}}{2} = 10\ \text{kHz}$$

Question 8: $BW_{\text{AM}} = 8\ \text{kHz}$; $BW_{\text{FM}} = 40\ \text{kHz}$; **Expansion Factor =** 5
Mathematical Derivation:

$$BW_{\text{AM}} = 2 \times 4\ \text{kHz} = 8\ \text{kHz}$$

$$BW_{\text{FM}} = 10 \times B_m = 10 \times 4\ \text{kHz} = 40\ \text{kHz}$$

$$\text{Expansion Factor} = \frac{BW_{\text{FM}}}{BW_{\text{AM}}} = \frac{40\ \text{kHz}}{8\ \text{kHz}} = 5$$

Question 9: $S = 64\ \text{kBaud};\ BW = 96\ \text{kHz}$
Mathematical Derivation: For binary signaling, $r = 1 \implies S = \frac{N}{r} = 64\ \text{kBaud}.$

$$BW = (1 + d) \times S = (1 + 0.5) \times 64 = 1.5 \times 64 = 96\ \text{kHz}$$

Question 10: **Bandwidth/direction =** $18\ \text{kHz};\ f_{c1} = 19\ \text{kHz};\ f_{c2} = 41\ \text{kHz}$
Mathematical Derivation: Total spectrum $= 50 - 10 = 40\ \text{kHz}$. Usable spectrum $= 40 - 4\ \text{kHz}\ (\text{guard}) = 36\ \text{kHz}.$

$$BW_{\text{each}} = \frac{36\ \text{kHz}}{2} = 18\ \text{kHz}$$

Forward span: $10\ \text{kHz}$ to $28\ \text{kHz} \implies f_{c1} = \frac{10+28}{2} = 19\ \text{kHz}.$
Guard span: $28\ \text{kHz}$ to $32\ \text{kHz}.$
Backward span: $32\ \text{kHz}$ to $50\ \text{kHz} \implies f_{c2} = \frac{32+50}{2} = 41\ \text{kHz}.$

Question 11: **[B] Noise primarily affects signal amplitude; an FM receiver can clip amplitude fluctuations without losing information encoded in frequency**
Explanation: Atmospheric and electromagnetic noise directly couples additively into amplitude. In FM, information is stored entirely in zero crossings (frequency shifts), so amplitude limiters can suppress noise spikes before demodulation.

Question 12: **1: Amplitude | 2: Frequency | 3: Phase | 4: Carrier Amplitude (switched discretely)**
Explanation: In analog modulations (AM, FM, PM), continuous variation alters amplitude, frequency, or phase. In ASK, digital bit patterns switch the carrier amplitude between discrete levels (e.g., on/off keying)[cite: 5].
